Answers to DICK questions:

1. What is the name of the course you are taking this test from?
   Distributed Industrial Computer Systems
2. Main objectives of Industry 4.0 are:
   Factories digitization
   Industrial IoT
   Cybersecurity
   Systems integration
3. What models of industrial computer systems you know?
   ISA95
   RAMI4.0
   Diabolo
   IIRA i IIoT
   IEC62264 based
4. What is SCADA?
   Supervisory Control and Data Acquisition
5. What are used to gather information from and deliver information to the object?
   Sensors and actuators
6. Which industrial revolution we are about to face now?
   4
7. Can there be 'buckets & pots' in the industrial networks?
   False
8. What areas of computer-oriented technology are involved in Control Systems?
   IT & OT
   Mobile Technologies & ICT
9. What are the characteristics of a programmable industrial controller?
   Real-time operations
   Environmental resistance
   Time control of program execution
10. Can you use SFC commands to change the bit value?
    False
11. Can there be holes in industrial networks?
    True
12. Can the industrial controllers be programmed by graphic diagrams?
    True

13. Can ICS be distributed?
    True
14. Can ICS be centralized?
    True
15. Transmission speed is the main feature of RT communication.
    False
16. How does ICS interact with physical objects?
    By nodes' inputs/outputs in control loops
    By data acquisition, control and regulation
17. Can the industrial controllers be programmed using graphic images?
    False
18. What types of Real-Time systems you know?
    Soft RTS
    Firm RTS
    Hard RTS
19. What industrial communication primitives do you know?
    Response
    Request
    Indication
    Confirmation
20. What are the known models of medium control?
    TDMA
    Token Passing
    Master-Slave
    PDC
21. What is characteristic of ICS activities?
    Real time data processing
    Time determinism
    Real time communication
    High speed networking
    Determinism
22. Which topology is used with fieldbuses?
    bus
23. What features are important for ICS?
    scalability
    flexibility
    accessibility

24. When designing an ICS, resistance to what factors should be considered?
    conductive disturbances
    inductive disturbances
25. What types of transactions are used in RT industrial networks?
    Cyclic scheduled
    Cyclic synchronous
    Acyclic (triggered)
26. What soft skills do you consider important in engineering activities with ICS?
    ability to talk and listen to other people
    think creatively
27. How does ICS nodes interact with physical objects?
    via computer networks (if interfaces available)
    by IO interfaces
28. What common operational issues are considered by industrial system models?
    development and maintenance
    product lifecycle
    business viewpoints
    systems integration and communication
29. What are the problems with media access control in industrial networks?
    Physical collisions
    Logical collisions
30. How does the controller check the program execution time?
    by watchdog(s)
31. What types of programming languages are used for industrial controllers?
    Graphic and text ones of low and high abstractions
32. What are the typical elements of a controller cycle?
    output control, input data acquisition, data processing
33. What types of network traffic in relation to time do you know?
    Real-Time (RT)
    Isochronous Real-Time (IRT)
    Non Real-Time (NRT)
    Time-Oriented Preemptive Flow (TOPF)